

L09 — Speech Synthesis II

Acoustic models, vocoders, style transfer

Study summary generated from lecture slides. ETH AI for Digital Characters.

Modern neural TTS

- Text analysis -> Acoustic model (mel spectrogram) -> Vocoder (waveform)

Tacotron 2

- Encoder: char embedding + conv layers + bi-LSTM
- Autoregressive decoder with attention -> mel frames
- Postnet refines mel spectrogram
- Pros: high quality; Cons: slow, attention instability (skip/repeat)

FastSpeech / FastSpeech 2

- Non-autoregressive: duration predictor + length regulator
- Parallel decoding -> fast inference, stable output
- FastSpeech 2: variance predictors for duration, energy, pitch
- Speed control: scale duration D (2xD -> half speed)
- Uni-TTS: multilingual FastSpeech 2 base (Microsoft Azure)

Vocoders

- Griffin-Lim: iterative phase from magnitude, no training, robotic
- WaveNet: autoregressive dilated causal convolutions, high quality, slow
- HiFi-GAN: GAN with multi-scale + multi-period discriminators, real-time
- DiffWave: diffusion-based, high quality, slow inference
- HiFi-GAN losses: adversarial + feature matching + L1 mel loss

Style / speaker control

- One-hot speaker labels vs learned speaker embeddings
- Embeddings enable interpolation and zero-shot adaptation
- Style tokens: unsupervised style discovery

Evaluation

- MOS: 1-5 naturalness rating; A/B preference tests
- DRT: 96 rhyming pairs in carrier phrases
- MRT: 50 sets of 6 similar words
- WER via ASR on synthesized speech (intelligibility)